

Oleksiy Nedobiychuk

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 Pisa, Italy

Education

- **M.Sc. in Computer Science (AI curriculum), University of Pisa** [2023 – Present] 109/110(e).
- **B.Sc. in Computer Science, University of Pisa** [2019 – 2023] 106/110.

Relevant Experience

Software Engineer at TD Group Italia S.r.l., Vecchiano, Italy [Jul. 2025 – Present]:

- Developed full-stack **Django/MySQL** platforms for agritech and healthcare IoT projects (VITE, Smart-Care, EyeDos), covering device management, role-based admin workflows, **MQTT** ingestion, and real-time updates with **Channels/Redis**.
- Built **Python** data pipelines and scheduled processors for sensor/weather data with **APScheduler**, **BeautifulSoup**, and **Requests**; improved grapevine models for phenology, maturation, irrigation, evapotranspiration, and disease-risk estimation.
- Created **Highcharts/ECharts** dashboards and data normalization tools combining **RapidFuzz**, **ONNX** embeddings, **HNSW** indexes, MySQL mapper tables, and **Keycloak**/OIDC role-based access.
- Built public **Django**/Tailwind project sites for HeartSentry and VITE, including content pages, contact/privacy flows, and deployable static/media configuration.

Full-Stack Developer at TAI Solutions S.r.l., Pisa, Italy [Sept. – Nov. 2022]:

- Built a **Spring** microservice and **Angular** application for tenant/user management, integrating **Keycloak** authentication and authorization across multiple clients.

Projects

Professional Cycling Analysis – Data Mining

[GitHub](#)

- Cleaned and engineered cycling datasets with imputation, outlier detection, and correlation analysis; applied K-means/DBSCAN clustering and trained a neural network to predict rider performance with 90% test accuracy.

Neural Network for Regression/Classification

[GitHub](#)

- Built a neural network from scratch in **Python/NumPy** with backpropagation, Nesterov momentum, L1/L2 regularization, and grid-search tuning; reached 0.01 MSE on regression and 95% accuracy on classification.

ProbLog Based AI Agent

[GitHub](#)

- Built a probabilistic AI agent for a 2D fighting game, modeling uncertainty with a ProbLog knowledge base and dynamic probability estimation; outperformed MCTS, MinMax, and Prolog baselines in simulation.

Web Application – Full-Stack Development

[Live](#)

- Developed and deployed a **Svelte**/WebGL2 frontend with a **Django**/PostgreSQL backend on Debian using Nginx and Apache, including database design and production server configuration.

Skills

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|---------------------|---|
| Programming | • Proficient: Python, TypeScript/JavaScript. Familiar: C++, C, Rust. Learning: Go. |
| Backend/Data | • Django, Spring, REST APIs, MQTT, PostgreSQL, MySQL, MongoDB, Redis. |
| Frontend | • Angular, Svelte, Highcharts, ECharts, WebGL2, responsive UI development. |
| ML/DS | • PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn. |
| Tools | • Linux (Ubuntu, Arch), Shell scripting, Git, Docker, Vim, Copilot, ChatGPT. |
| Languages | • Fluent in English, Italian, and Ukrainian. |